

Multiple Choice

1. **Answer:** A

Options: A) cause particles to become neutral; B) increase the mass of the nucleus; C) cause particles to spin; D) generate heat within the nucleus; E) create magnetic fields

Note: In the atomic nucleus, electrical forces cause positively charged protons to repel each other, while the presence of neutrons helps to neutralize these repulsive forces, contributing to the overall stability of the nucleus.

2. **Answer:** D

Options: A) 1000 coul; B) 1800 coul; C) 3000 coul; D) 1500 coul; E) 2500 coul

Note: The correct answer is 1500 coulombs because charge is calculated by multiplying current (in amperes) by time (in seconds), so 50 A for 30 seconds (1/2 min) yields 50 A $\times$ 30 s = 1500 C.

3. **Answer:** A

Options: A) mass; B) low-frequency sound; C) nuclear power; D) energy; E) heat

Note: The correct answer is based on Einstein's theory of relativity, which establishes a deep connection between electricity, magnetism, and mass, demonstrating that energy (from electric and magnetic fields) can be converted into mass, as expressed in the equation $E=mc^2$.

4. **Answer:** E

Options: A) 2 kT; B) (7/2) k T; C) 5 kT; D) (1/4) k T; E) 3 kT

Note: In a three-dimensional harmonic oscillator, each degree of freedom contributes an average energy of $\frac{1}{2}kT$ due to equipartition of energy, resulting in a total average energy of $3 \times \frac{1}{2}kT = \frac{3}{2}kT$ for kinetic energy and the same for potential energy, leading to an overall average total energy of $3kT$.

5. **Answer:** B

Options: A) -45 ft/sec² and 4 sec; B) -48.4 ft/sec² and 4.55 sec; C) -40 ft/sec² and 6 sec; D) -42 ft/sec² and 5.5 sec; E) -52 ft/sec² and 4.8 sec

Note: The correct answer is derived from applying the kinematic equations of motion, where the airplane's initial velocity, final velocity (zero upon stopping), and stopping distance allow for the calculation of acceleration as -48.4 ft/sec² and the time to stop as 4.55 seconds through the relationship $v^2 = u^2 + 2as$ and $v = u + at$.

True / False

6. **Answer:** False

Options: A) True; B) False

Note: The total energy required to remove both electrons from a helium atom is

approximately 79 eV, as the first ionization energy is about 24.6 eV and the second ionization energy is around 54.4 eV, leading to a total that exceeds 39.5 eV.

7. **Answer:** False

Options: A) True; B) False

Note: Solar radiation primarily influences Earth's surface temperature and atmospheric processes, while Earth's internal energy is mainly generated by radioactive decay and residual heat from its formation, not through thermal conduction and gravitational pressure.

8. **Answer:** False

Options: A) True; B) False

Note: The mass of an atom primarily comes from its protons and neutrons, as they have significantly greater mass than electrons, while the volume of an atom is mainly due to the space occupied by its electron cloud rather than the volume of the protons.

9. **Answer:** False

Options: A) True; B) False

Note: The statement is false because protons are currently understood to be stable particles with no confirmed decay, thus they do not have a half-life.

10. **Answer:** False

Options: A) True; B) False

Note: The statement is false because the power output of 33 horsepower, which is equivalent to approximately 24,500 watts, cannot produce a force of 300 lbs at a speed of 45 mi/hr, as the required power to maintain that force at that speed would exceed 33 horsepower.

Long Form Response

11. **Answer:** To analyze the motion of a stone thrown horizontally from a bridge with an initial speed of 10 m/s, we first consider the vertical motion under the influence of gravity. The stone falls 80 m vertically, and using the equation of motion $s = ut + \frac{1}{2}gt^2$ (where s is the vertical distance, u is the initial vertical speed, g is the acceleration due to gravity, and t is time), we set $u = 0$ since the initial vertical velocity is zero. Rearranging the equation for time yields $t = \sqrt{\frac{2s}{g}}$. Substituting $s = 80$ m and $g \approx 9.81$ m/s², we find $t \approx \sqrt{\frac{160}{9.81}} \approx 4$ seconds. Thus, the stone takes approximately 4 seconds to reach the water, confirming the key answer.
Note: correct because it appropriately applies the kinematic equation for vertical motion, recognizing that the stone's initial vertical speed is zero, allowing for the calculation of the time to fall 80 m solely under the influence of gravity.
12. **Answer:** To analyze the motion of passengers on a carnival ride moving in a horizontal circle with a radius of 5.0 m, completing a revolution in 4.0 s, we first calculate the linear velocity (v) using the formula $v = 2\pi r/T$, where r is the radius and T is the

period. Substituting the values gives $v = 2(5.0 \text{ m})/(4.0 \text{ s}) = 3.93 \text{ m/s}$. The centripetal acceleration (a_c) can then be calculated using the formula $a_c = v^2/r$, leading to $a_c = (3.93 \text{ m/s})^2/(5.0 \text{ m}) = 3.08 \text{ m/s}^2$. This acceleration indicates that a centripetal force must act upon the passengers, directed towards the center of the circular path, which is crucial for maintaining their circular motion. The implications of this acceleration in terms of safety suggest that the ride must be designed to ensure that the centripetal force is sufficient to keep passengers securely in their seats, preventing potential hazards associated with excessive acceleration or inadequate centripetal force.

Note: correct because it accurately applies the formulas for linear velocity and centripetal acceleration to determine the motion of passengers on a carnival ride, demonstrating the relationship between these calculations and the implications for centripetal force and safety.

End of Answer Key